

Arjun Menon

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Education

Princeton University

Princeton, New Jersey

B.S.E. in Computer Science

2023 - 2027

Junior Independent Research: Toward a Principled Recipe for VLA Distillation

- Advisor: Dhruv Shah

Advisors: Dhruv Shah, Tri Dao, Tom Griffiths

Graduate Courses: Computer Vision; Behavioral Imaging; Data-Driven Robotics; Information Theory; Machine Behavior; Multi-Robot Systems; Networking for AI; Optimization for ML; Reinforcement Learning; Safety-Critical Robotics; Systems and ML; Theoretical ML; Theoretical Deep Weakly Supervised Learning; Transformers and LLMs

Research Interests

Embodied Reasoning

Multimodal Models - Efficient Training/Inference, Model Architecture

Reasoning - Cognitive Priors, Policy Generation

World Modeling - Latent Representation, Embedding Geometry

Publications

1. Han Guo, Jack Zhang, **Arjun Menon**, Driss Guessous, Vijay Thakkar, Yoon Kim, and Tri Dao. "CODA: Rewriting Transformer Blocks as GEMM-Epilogue Programs." In *Advances in Neural Information Processing Systems 39 (NeurIPS 2026)*, 2026. Poster.
2. **Arjun Menon**, Dilip Arumugam, and Thomas L. Griffiths. "Posterior Recovery for Reinforcement Learning." Manuscript under review, 2026.
3. Yuzheng Xu, Annya Dahmani, Matthew D. Blanchard, Niclas Dern, Edy Nastase, Francesca Bianco, Maja Pavlovic, Sukanya Krishna, Eric Modesitt, Miranda Anna Christ, Arth Singh, Gaia Molinaro, Sikata Bela Sengupta, Jaji Pamarthi, **Arjun Menon**, and Rishub Jain. "Toward Human-AI Complementarity Across Diverse Tasks." *arXiv preprint arXiv:2605.04070*, 2026.
4. Wang Yang*, Hongxuan Liu*, Xinghui Xu*, **Arjun Menon***, Xiaoran Cai*, Yunyu He*, Jingzong Zhou, Mengzhong Ma, Xinpeng Wei, Nathaniel Dennler, Yi Yu, Shaobo Wang, Cheng Peng, Aoran Jiao, Alexei Korolev, Ashis G. Banerjee, Yanyan Zhang, Kai Ye, Xinpeng Li, Chengquan Guo, Jingjing Fu, Marius Urbonas, Traian Tus, Gaoyue Zhou, George Ortiz, Irmak Guzey, Silei Ren, Lars Johansmeier, Rohit Sharma, Felix Feng, Yoshua Bengio, and Peng Qi. "Towards Trustworthy Physical AI: From Theory to Practice Across Life Cycle." *arXiv preprint arXiv:2607.22877*, 2026.
*Equal contribution.
5. Kia Ghods, Patty Liu, Katerina Labrou, Kincaid MacDonald, **Arjun Menon**, and Addison Wu. "Evidence Against LLM Homogenization in Creative Writing." Preprint, 2025. Available at: <https://a-memon.github.io/assets/pdf/LLMHomogenization.pdf>.
6. **Arjun Menon** and David Beloglazov. "Modular Counts, Smarter Programs: Enhancing Transformer Program Generalization with Structured Biases." Preprint, 2025. Available at: <https://a-memon.github.io/assets/pdf/BiasedTransformerPrograms.pdf>.
7. Erich Liang, Hang Pham, Sabrina Van, and **Arjun Menon**. "RotoSLAM: Improving SLAM Scale Consistency for Rotational Motion." Preprint, 2024. Available at: <https://a-memon.github.io/assets/pdf/RotoSLAM.pdf>.

Research Experience

1/26 - Present | Princeton Robotic Intelligence and Systems Lab (PRISM) | Researcher

Advisor: Dhruv Shah

○ **VLA Distillation |**

Designed and executed a comprehensive VLA distillation study across π - and MolmoBot-family models, training compressed policies from scratch and via LoRA finetuning; derived the optimal distillation recipe through controlled ablations and validated it across MolmoSpaces, DROID, and PolaRiS.

Built an end-to-end OpenPI stack for data generation, teacher relabeling, distributed training, and closed-loop evaluation; constructed expert trajectories, teacher-controlled trajectories, and teacher labels on expert-visited states to isolate the effects of action supervision and state distribution.

○ **VLA Efficiency |**

Profiled SOTA VLA inference and identified the action expert as the dominant latency contributor, contrary to prevailing VLM-focused assumptions; matched BF16 performance with W8 and action-sensitive W4 quantization and reduced projection time by ~40% with cached and fused action-expert kernels.

○ **FlowMap |**

Independently developed FlowMap, a behavior-sensitive depth-compression method across the vision-language backbone and action expert, with GPU profiling, distributed fitting, calibration, and closed-loop evaluation.

5/25 - Present | Princeton Dao AI Lab | Researcher

Advisor: Tri Dao

○ **CODA |**

Co-developed CODA, an LLM-driven GPU-kernel system that rewrites memory-bound Transformer computation as GEMM-epilogue programs; designed LLM-facing program abstractions, compiler-lowering rules, and hierarchical ASTs for kernel generation.

11/24 - Present | Princeton Computational Cognitive Science Lab | Researcher

Advisor: Tom Griffiths

○ **Posterior Recovery for Reinforcement Learning |**

Developed a Bayesian RL method inspired by failure-triggered belief revision, detecting failed task hypotheses mid-episode and updating or resampling the posterior within the same trajectory; generalized method across deterministic, stochastic, and combinatorial dynamics.

9/25 - 1/26 | Supervised Program for Alignment Research | Mentee

Mentor: Rishub Jain

○ **Human-AI Complementarity |**

Implemented data processing, methods, and evaluation for a 1,886-example study of human-AI complementarity across diverse tasks; explored self-confidence and information-theoretic routing criteria for hybrid human-AI decision making.

11/24 - 5/25 | Princeton Natural and Artificial Minds | Researcher

○ **Mechanistic Interpretability |**

Explored Bayesian and GFlowNet-based approaches to task-specific Transformer-head ablation and organized the item-property taxonomy used to characterize functional structure across components.

4/24 - 12/24 | Princeton Vision & Learning Lab | Researcher

○ **Infinigen |**

Replaced greedy object placement in Infinigen with multi-objective simulated annealing across successive procedural-generation stages.

Projects

5/26 | Exploring the Effectiveness of Geometrically Structured Representations for Visual Imitation

- Showed that temporally structured visual representations improve imitation learning on difficult manipulation tasks, and that simple Gaussian structure in the latent space provides strong signals for policy failure, OOD detection, and recovery.

11/25 | Looking for Trouble: Online Safe Exploration with Learned Dynamics Models

- Developed a safe-exploration method that replaces assumed global smoothness with a learned probabilistic dynamics model, using model uncertainty for safe-set expansion with finite-sample and infinite-horizon probabilistic safety guarantees.

7/25 | Profile-Kernel

- Built a CLI tool for NVIDIA Nsight Compute to pinpoint CUDA kernel bottlenecks from source code; reached 1K+ PyPI downloads.

7/25 | Evidence Against LLM Homogenization in Creative Writing

- Tested claims of LLM idea homogenization across embedding, compression, and semantic-diversity measures with context- and output-length controls, finding that classical evaluation metrics are inadequate indicators of semantic homogeneity.

5/25 | Modular Counts, Smarter Programs: Enhancing Transformer Program Generalization with Structured

Biases

- Added counting, positional structure, and modular routing to Transformer Programs, raising OOD accuracy from 50.9% to 98.1%.

12/24 | Video Reverse-Engineering in Frontier Foundation Models: Are We Getting There?

- Evaluated frontier multimodal models against task-specific neural baselines for detecting manipulation and tampering in short-form video; released the resulting analysis as a technical white paper.

5/24 | RotoSLAM: Improving SLAM Scale Consistency for Rotational Motion

- Generated synthetic trajectories for training and evaluating diffusion-depth supervision designed to reduce monocular SLAM scale inconsistency under near-pure camera rotation.

Honors and Awards

2025 | Anthropic AI Alignment Hackathon | **3rd Place Overall**

- **1 of 40** invited nationally

2024 | Palantir Winter Defense Tech Fellowship | **Fellow**

2024 | UChicago Trading Competition | **2nd Place Overall**

- **1 of 4** selected from Princeton as a freshman

2023 | William Lowell Putnam Mathematical Competition | **Top 700**

- 22/120; 3rd-highest Princeton Class of 2027 scorer and highest-scoring Princeton participant under 20